Spring potential energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate stored spring energy and distinguish it from force.

SUCCESS CRITERIA - I can...

1. Use displacement in meters.
2. Apply the squared relationship for spring energy.
3. Explain how stored energy can become motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart starts from rest at height 3 m. Can it reach 5 m with no external energy input or losses? Justify using kinetic energy.

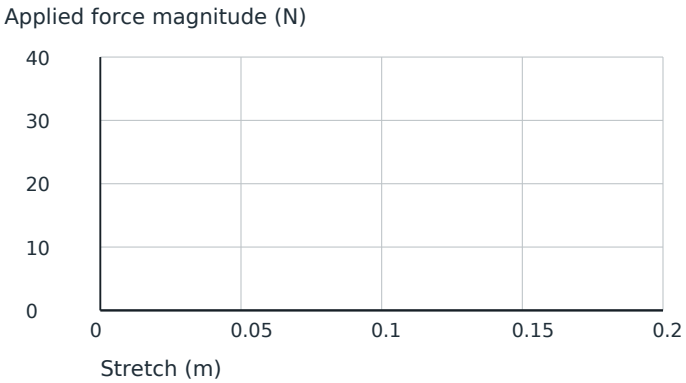
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $U_s = \frac{1}{2}kx^2$; $|F_{spring}| = k|x|$.
s means spring. Compression and stretch both store energy relative to the natural length; energy and force are different quantities.

Worked reasoning: A spring with k=200 N/m stretched 0.10 m stores $\frac{1}{2}kx^2$=1 J. Doubling stretch to 0.20 m stores 4 J, not 2 J.

Your turn: Draw force-versus-stretch triangles and compare their areas.



Marzano check - circle: ① ② ③

03 Read and interpret the model

The same spring is displaced x and 2 x. Compare stored energies and force magnitudes.



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: ① ② ③

04 Apply the physics

A spring launcher has k = 200 N/m and compression 0.10 m. It launches a 0.50 kg cart along a frictionless horizontal track. Label the spring model. Calculate stored energy and the cart's speed when the spring reaches its natural length.



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why is energy positive for both positive and negative x even though spring force reverses?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 8 N force acts at 120° to a 6 m displacement. Calculate its work and explain the sign.
Label force and displacement, including their relative direction.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

07 Exit check

k=80 N/m; stored energy 0.40 J. Find displacement magnitude and force magnitude.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.5 kg cart is launched by a spring k=400 N/m compressed 0.15 m with no losses. Find release speed. What compression doubles that speed?



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3